

Q11(b). Branch Prediction – Simple & Easy Explanation (11 Marks)

1. Motivation (Why Branch Prediction?)

In pipelining, a branch (like if/else or loops) creates uncertainty.

If the CPU waits to know the actual path, the pipeline stalls or gets flushed.

Branch prediction guesses the path early so the pipeline keeps running smoothly.

2. Static Branch Prediction (Fixed rules – no history)

(a) Always Taken – CPU assumes every branch is taken.

(b) Always Not Taken – CPU assumes branch is not taken.

(c) Backward Taken, Forward Not Taken (BTFNT):

- Backward branches (loops) → Taken
- Forward branches → Not Taken

Advantages:

- Very simple
- Low hardware cost
- Works well for simple pipelines

3. Dynamic Branch Prediction (Uses runtime behavior)

Dynamic prediction learns from past outcomes.

3.1 One-Bit Predictor:

- Stores the last outcome (Taken/Not Taken) for each branch.
- Predicts the same behavior next time.

Problem:

- Fails at loop exits because one mistake flips the prediction.

3.2 Two-Bit Saturating Counter:

4 states:

1. Strongly Taken
2. Weakly Taken
3. Weakly Not Taken
4. Strongly Not Taken

- Needs two mistakes to switch strong states → more stable.

Simple Diagram of States:

Strong NT $\leftrightarrow$ Weak NT $\leftrightarrow$ Weak T $\leftrightarrow$ Strong T

3.3 Local History Predictor:

- Keeps separate history for each branch.
- Works well when each branch has its own pattern.

3.4 Global History Predictor:

- Stores outcomes of recent branches in a Global History Register (GHR).
- Predicts patterns involving multiple branches.

3.5 Two-Level / Tournament Predictor:

- Combines local and global predictors.
- Selector chooses the best-performing predictor.
- Used in modern processors.

4. Simple Diagrams

(a) Pattern History Table (PHT):

PC $\rightarrow$ [Pattern History Table] $\rightarrow$ Prediction

(1-bit or 2-bit counters)

(b) Two-Bit Counter State Diagram:

Strong T $\leftrightarrow$ Weak T $\leftrightarrow$ Weak NT $\leftrightarrow$ Strong NT

5. Conclusion

Branch prediction keeps pipelines full by guessing outcomes early.

Static prediction is simple.

Dynamic prediction is smarter because it learns from runtime behavior and gives higher accuracy.